Service Tool Instruction

Service Tool Instruction Number: 3377905-1	Revision Level:	Released Date: 27-jan-2004

Description

Thread Insert Kit

Purpose

This document provides information for repairing damaged threads using thread insert kit, Part Number 3376448. The kit is used with alignment fixture, Part Number ST 1232.

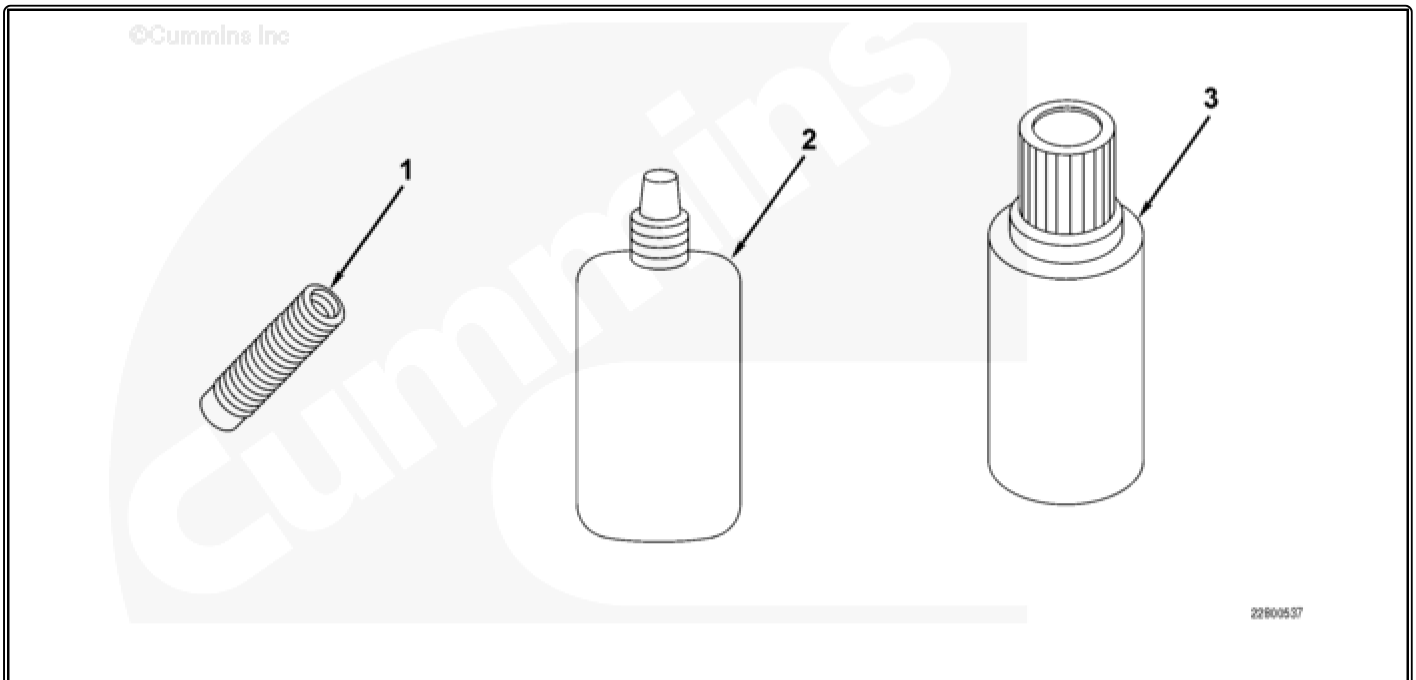
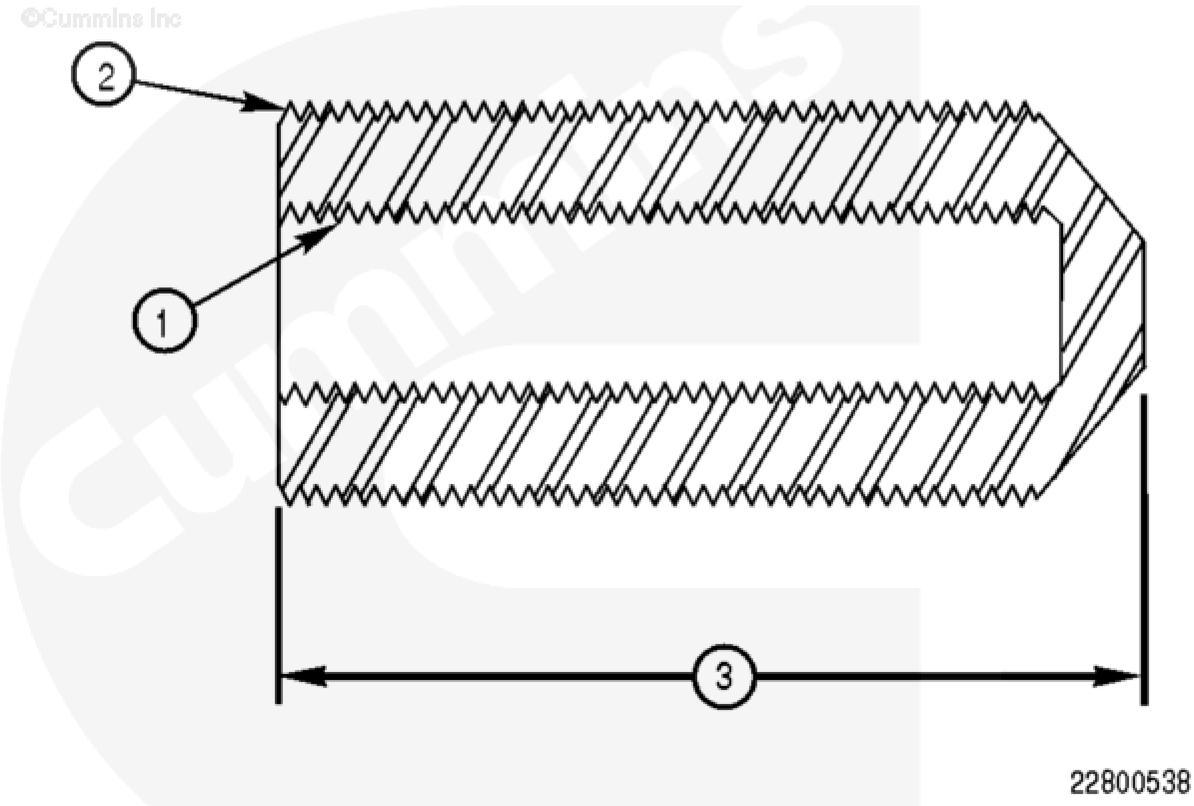


Table 1, Thread Insert Kit, Part Number 3376448			
Item Number	**Part Number**	**Description**	**Quantity**
1	3376447	Insert, 9/16-18 (1 shown)	10
1	3376437	Insert, 1/4-20 (1 shown)	10
1	3376438	Insert, 5/16-18 (1 shown)	10
1	3376439	Insert, 3/8-16 (1 shown)	20
1	3376440	Insert, 3/8-24 (1 shown)	20

Table 1, Thread Insert Kit, Part Number 3376448			
Item Number	**Part Number**	**Description**	**Quantity**
1	3376441	Insert, 7/16-14 (1 shown)	10
1	3376442	Insert, 7/16-20 (1 shown)	15
1	3376443	Insert, 1/2-13 (1 shown)	15
1	3376444	Insert, 5/8-11 (1 shown)	10
1	3376445	Insert, 5/8-18 (1 shown)	10
1	3376446	Insert, 1/2-20 (1 shown)	10
2	3824038	Thread-locking compound	1
3	3824715	Primer	1
Not shown	3163757	Storage case	1
Not shown	3822708	Kit content bulletin	1

Table 2, Items Used with Thread Insert Kit, Part Number 3376448, Purchased Separately			
Item Number	Part Number	Description	Quantity
Not shown	ST 1232	Drill/ream fixture	1
Not shown	3164019	Chip vacuum	1
Not shown	See Table 3	Drill bushing	1



Insert

Table 3, Thread Insert Dimensions					
Part Number	Internal Thread Size (1)	External Thread Size (2)	Drill Size	Overall Length (3)	Drill Bushing
3376347	9/16-18	11/16-16	33/64 (0.609)	1.18-in	ST-1234
3376437	1/4-20	3/8-16	5/16 (0.312)	0.75-in	4918170
3376438	5/16-18	7/16-14	U (0.368)	0.93-in	4918171
3376439	3/8-16	1/2-13	27/64 (0.421)	1.13-in	4918172
3376440	3/8-24	1/2-13	27/64 (0.421)	1.07-in	4918172
3376441	7/16-14	9/16-12	31/64 (0.484)	1.15-in	3376495
3376442	7/16-20	9/16-12	31/64 (0.484)	1.21-in	3376495
3376443	1/2-13	5/8-11	17/32 (0.531)	1.51-in	3376495
3376444	5/8-11	13/16-16	3/4 (0.750)	1.64-in	ST-1237

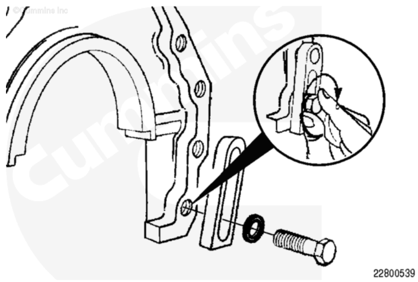
Table 3, Thread Insert Dimensions

Part Number	Internal Thread Size (1)	External Thread Size (2)	Drill Size	Overall Length (3)	Drill Bushing
3376445	5/8-18	13/16-16	3/4 (0.750)	1.36-in	ST-1237
3376446	1/2-20	5/8-11	17/32 (0.531)	1.19-in	3376495

Install the correct spacers and washers onto the fixture alignment capscrews.

Install the capscrew and spacer assemblies through the mounting slot of the fixture.

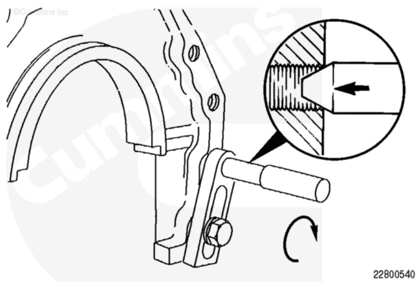
Finger-tighten the fixture mounting capscrew.



Use a tapered pin to align the guide bar and bushing over the damaged capscrew thread.

Push downward on the tapered pin and tighten the fixture mounting capscrew.

Torque Value: 67 n•m [49 ft-lb]



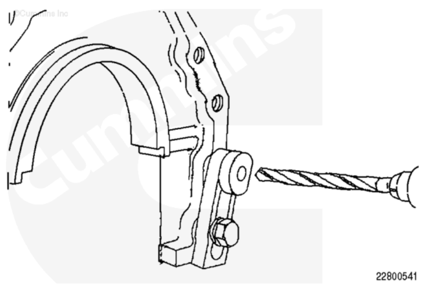
Using the information in Table 3, obtain the correct drill size.

Install the appropriate size drill/ream bushing.

If necessary, mark the drill with tape for the required drill depth before starting the drilling operation.

Drill to the bottom of the original capscrew hole.

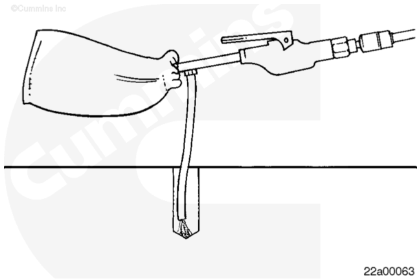
Remove the alignment fixture.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use the chip vacuum attached to compressed air to remove the chips from the newly drilled capscrew hole.

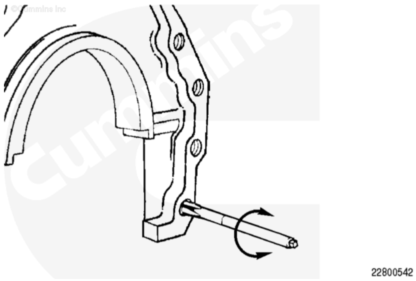


Note : Stop frequently to clean the bore using a chip vacuum.

Using the information contained in Table 3, External Thread Size, obtain correct tap size.

Tap the full depth of the hole using a suitable tapping fluid.

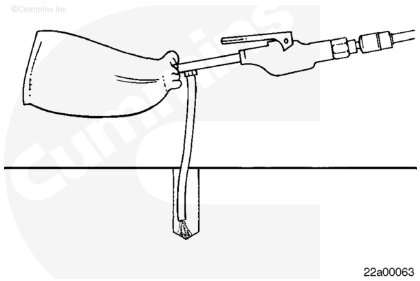
Remove the tap.



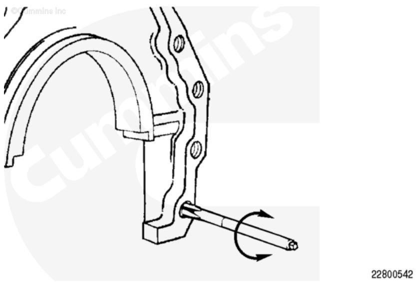
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use the chip vacuum attached to compressed air to remove the chips from the newly tapped capscrew hole.

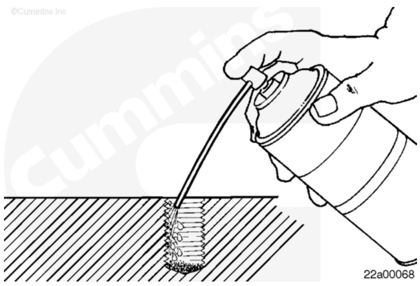


Thread the appropriate tap in and out of the newly cut threads several times to verify the thread condition, and clean debris from the threads.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Note : Check the capscrew hole for evidence of porosity and cracks.

Clean the threads, and flush any debris from the tapped hole using a solvent-degreasing agent.

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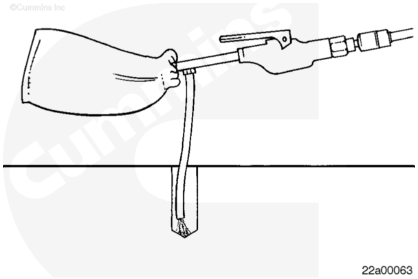
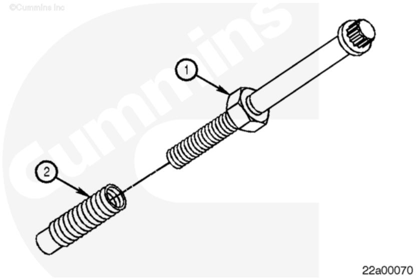


Use the chip vacuum attached to compressed air to remove the chips and solvent-degreasing agent from the newly tapped capscrew hole.

Place the appropriate jam nut (1) on a capscrew.



Install the appropriate thread salvage insert (2) onto the capscrew until the capscrew nears the bottom of the insert. Rotate the jam nut (1) until it contacts the thread salvage insert (2).

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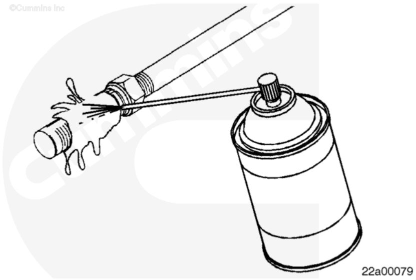

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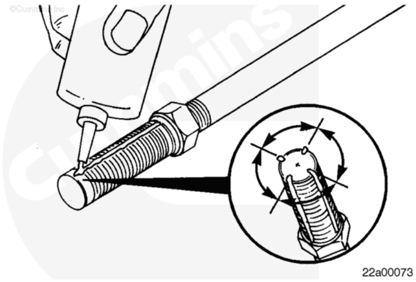
Remove the preservative coating from the circumference of the thread salvage insert using a solvent-degreasing agent.

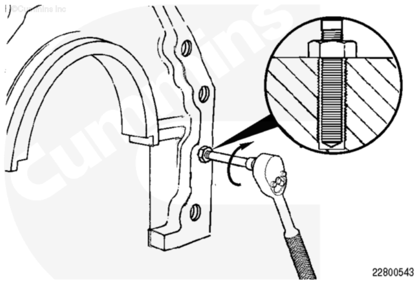
Apply a light coat of Primer to the thread insert and the threaded hole; allow 3 to 5 minutes to dry.

Apply four beads of thread-locking compound, Part Number 3824038, to the circumference of the thread salvage insert. Each bead **must** be 0.8 mm [0.03 in] wide down the entire length of the thread salvage insert, and the beads **must** be 90 degrees apart.



Install the insert until it is flush with the block or it bottoms in the hole.

Allow to dry for 3 hours.



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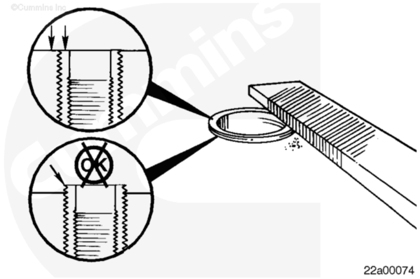
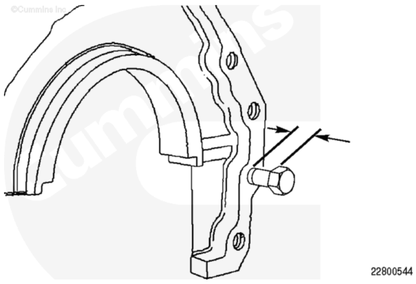


Remove the capscrew and jam nut.

If necessary, file or machine the top of the thread salvage insert so that it is flush with the surface.

Make certain that the inner diameter of the thread salvage insert is free of burrs. Remove any file shavings or debris with the chip vacuum.

Note : It is important that sufficient thread engagement be maintained after this repair.



Install a replacement capscrew until it bottoms out in the repaired hole.

Measure the distance from under the capscrew head to the component's surface. Make certain that this dimension is a minimum of 2 mm [0.079 in] less than the thickness of the component and gasket to be attached.

If the distance is within tolerance, the installation is complete. If the distance is **not** within tolerance, the capscrew can be shortened slightly, or a shorter capscrew can be used. If a shorter capscrew is used, make certain that a minimum thread engagement of 1-1/2 times the capscrew diameter is maintained.

Last Modified: 02-Feb-2004